# INFORME DE LABORATORIO

## (formato estudiante)

| INFORMACIÓN BÁSICA | | | | | |
|---|---|---|---|---|---|
| **ASIGNATURA:** | Programacion Web 2 | | | | |
| **TÍTULO DE LA PRÁCTICA:** | Python | | | | |
| **NÚMERO DE PRÁCTICA:** | 04 | **AÑO LECTIVO:** | 2026-B | **NRO. SEMESTRE:** | *II* |
| **FECHA DE PRESENTACIÓN** | 13/10/2024 | **HORA DE PRESENTACIÓN** | *15:10* | | |
| **INTEGRANTE (s):**<br><br>Tapia Huamantuma Antony Juan | | | | **NOTA:** | |
| **DOCENTE(s):**<br>Lino Pinto | | | | | |

## SOLUCIÓN Y RESULTADOS

### I. SOLUCIÓN DE EJERCICIOS/PROBLEMAS

*Chespictures.py:*

```
from pieces import *
from picture import *

bishop = Picture(BISHOP)
king = Picture(KING)
knight = Picture(KNIGHT)
pawn = Picture(PAWN)
queen = Picture(QUEEN)
rock = Picture(ROCK)
square = Picture(SQUARE)
```

*colors.py*

```
WHITE = (255, 255, 255)
BLACK = (0, 0, 0)
LIGHTGRAY = (200, 200, 200)
GRAY = (127, 127, 127)
DARKGRAY = (50, 50, 50)
BLUE = (0, 0, 255)

color = {
   '_': LIGHTGRAY,
```

```
  '=': GRAY,
  '.': WHITE,
  '@': BLACK,
  '#': DARKGRAY,
  ' ': BLUE,
}
inverter = {
  '_': '=',
  '=': '_',
  '.': '@',
  '@': '.',
}
```

*Interpreter.py*

```
import pygame, sys
from pygame.locals import *
from colors import *

def parseLine(DISPLAY, y, s):
  x = 0
  for c in s:
    pygame.draw.line(DISPLAY, color[c], (x, y), (x, y))
    x += 1

def draw(picture):
  try:
    img = picture.img
  except:
    img = picture
  pygame.init()

  DISPLAY=pygame.display.set_mode((640, 480))
  DISPLAY.fill(BLUE)

  n = len(img)
  for i in range(0, n):
    parseLine(DISPLAY, i, img[i])

  while True:
    for event in pygame.event.get():
      if event.type==QUIT:
        pygame.quit()
        sys.exit()
    pygame.display.update()
```

*picture.py*

```
from colors import *
```

```
class Picture:
  def __init__(self, imagen):
    self.imagen = imagen

  def __eq__(self, otro):
    return self.imagen == otro.imagen

  def _invColor(self, rgb):
    if rgb not in inverter:
      return rgb
    return inverter[rgb]

  def verticalMirror(self):
    arriba = []
    for valor in self.imagen:
      arriba.append(valor[::-1])
    return Picture(arriba)

  def negative(self):
    Otraimagen = []
    for valor in self.img:
      row = []
      for char in valor:
        row.append(self._invColor(char))
      Otraimagen.append(row)
    return Picture(Otraimagen)

  def join(self, p):
    Otraimagen = []
    for variable, valor in enumerate(self.img):
      Otraimagen.append(list(valor) + list(p.img[variable]))
    return Picture(Otraimagen)

  def up(self, p):
    Otraimagen = []
    for valor in p.img:
      Otraimagen.append(valor[::1])
    for valor in self.img:
      Otraimagen.append(valor[::1])
    return Picture(Otraimagen)

  def under(self, p):
    Otraimagen = []
    for valor in self.img:
```

```
        Otraimagen.append(list(valor))
    for i, valor in enumerate(p.img):
      for j, char in enumerate(valor):
        if(Otraimagen[i][j] == ' '):
          Otraimagen[i][j] = char
    return Picture(Otraimagen)

  def horizontalRepeat(self, n):
    auxiliar = self
    for _ in range(n-1):
      auxiliar = auxiliar.join(self)
    return auxiliar

  def verticalRepeat(self, n):
    auxiliar = self
    for _ in range(n-1):
      auxiliar = auxiliar.up(self)
    return auxiliar
```

*pieces.py*

```
SQUARE = [
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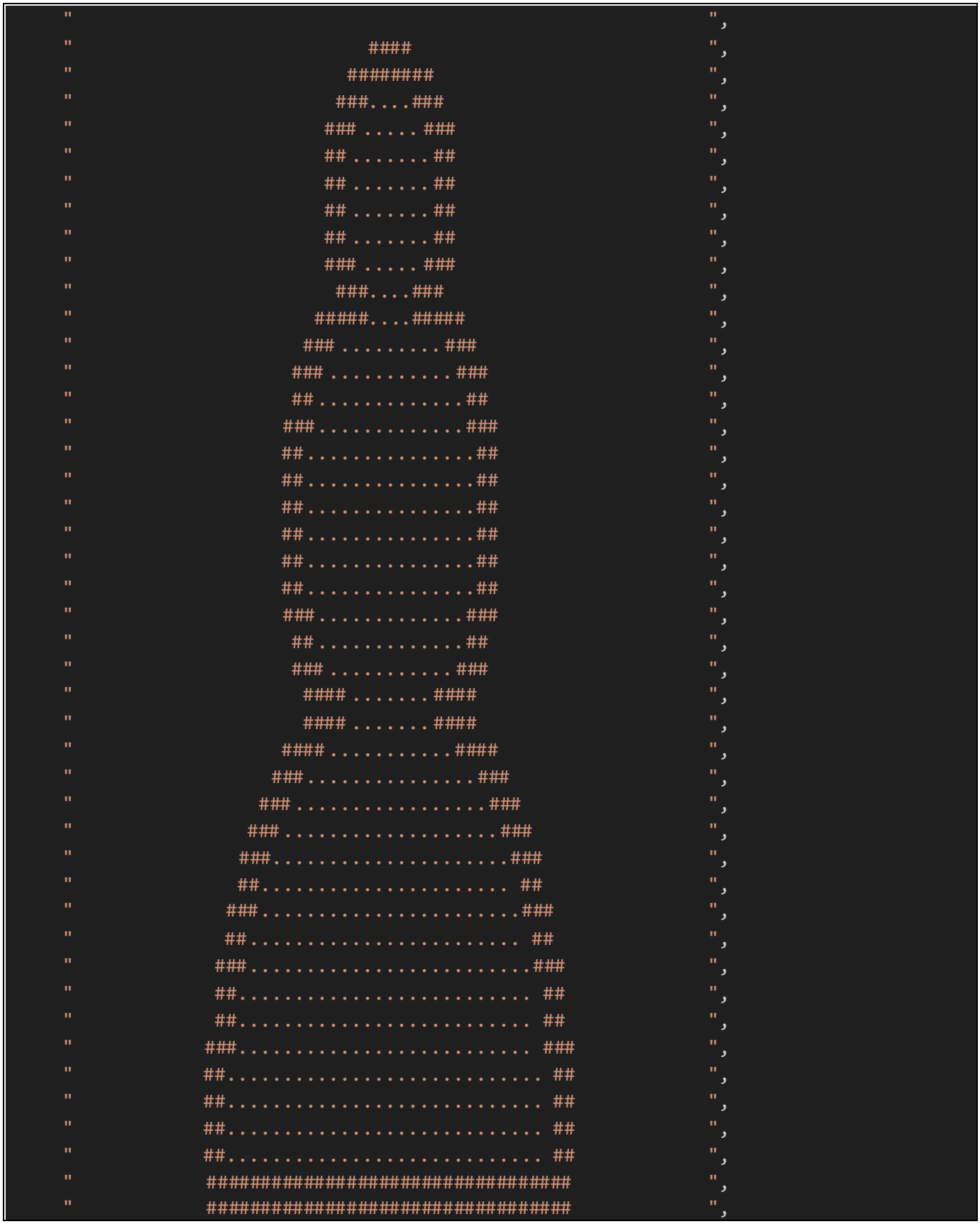
KING = [
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KNIGHT = [
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PAWN = [
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QUEEN = [
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```

*Test.py*

```
from chessPictures import *
from interpreter import draw
from picture import Picture
from colors import *

#Ejercicio a
```

```
tab = knight
tab = Picture.join(tab, Picture.negative(knight))
tab = Picture.up(Picture.negative(tab), tab)
draw(tab)


#Ejercicio b

tab = knight
tab = Picture.join(tab, Picture.negative(knight))
tab = Picture.up(Picture.verticalMirror(tab), tab)
draw(tab)


#Ejercicio c

tab = queen
tab = Picture.horizontalRepeat(queen, 4)
draw(tab)


#Ejercicio d

tab = square
tab = Picture.join(tab, Picture.negative(square))
tab = Picture.horizontalRepeat(tab, 4)
draw(tab)


#Ejercicio e

tab = square
tab = Picture.join(tab, Picture.negative(square))
tab = Picture.negative(Picture.horizontalRepeat(tab, 4))
draw(tab)


#Ejercicio f

tab1 = square
tab1 = Picture.join(tab1, Picture.negative(square))
tab1 = (Picture.horizontalRepeat(tab1, 4))
tab2 = Picture.negative(tab1)
tab3 = Picture.verticalRepeat(Picture.up(tab2, tab1), 2)
draw(tab3)
```

```
#Ejercicio g

filaInit = Picture.under(rock, Picture.negative(square))

filaInit = Picture.join(filaInit, Picture.under(knight, square))
filaInit = Picture.join(filaInit, Picture.under(bishop, Picture.negative(square)))
filaInit = Picture.join(filaInit, Picture.under(queen, square))
filaInit = Picture.join(filaInit, Picture.under(king, Picture.negative(square)))
filaInit = Picture.join(filaInit, Picture.under(bishop, square))
filaInit = Picture.join(filaInit, Picture.under(knight, Picture.negative(square)))
filaInit = Picture.join(filaInit, Picture.under(rock, square))

filaNegro = Picture.negative(filaInit)

filaPeon = Picture.under(pawn, square)
filaPeon = Picture.join(filaPeon, Picture.under(pawn, Picture.negative(square)))
filaPeon = Picture.horizontalRepeat(filaPeon, 4)

tab1 = square
tab1 = Picture.join(tab1, Picture.negative(square))
tab1 = (Picture.horizontalRepeat(tab1, 4))
tab2 = Picture.negative(tab1)
tab3 = Picture.verticalRepeat(Picture.up(tab2, tab1), 2)

tablero = Picture.up(Picture.negative(filaPeon), filaNegro)
tablero = Picture.up(tab3, tablero)
tablero = Picture.up(filaPeon, tablero)
tablero = Picture.up(filaInit, tablero)

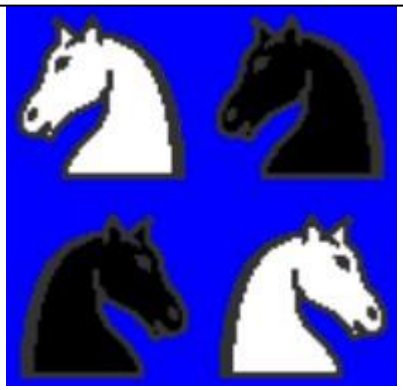
draw(tablero)
```

*Resolucion problemas:*

*a.*



*b.*



*c.*



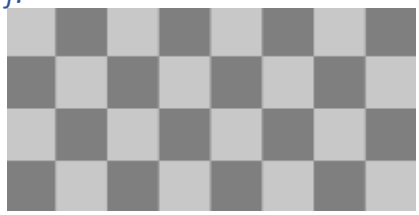
*d.*



*e.*



*f.*



*g.*



## REFERENCIAS Y BIBLIOGRAFÍA

- https://www.w3schools.com/python/python_reference.asp
- https://docs.python.org/3/tutorial/´

| Contenido y demostración | | Puntos | | Checklist Estudiante | |
|---|---|---|---|---|---|
| | | | | Profesor | |
| 1. GitHub | Repositorio se pudo clonar y se evidencia la estructura adecuada para revisar los entrega bles. (Se descontará puntos por error o onser vación) | 4 | | | |
| 2. Commits | Hay porciones de código fuente asociado a los commits planificados con explicaciones deta lladas. (El profesor puede preguntar para re f rendar calificación). | 2 | | | |
| 3. Ejecución | Se incluyen comandos para ejecuciones y pruebas del código fuente explicadas gradualmente que permitirían replicar el proyecto. (Se descontará puntos por cada omisión) | 2 | | | |
| 4. Pregunta | Se responde con completitud a la pregunta formulada en la tarea. (El profesor puede preguntar para refrendar calificación). | 1 | | | |
| 7. Ortografía | El documento no muestra errores ortográficos. (Se descontará puntos por error encontrado) | 2 | | | |
| 8. Madurez | El Informe muestra de manera general una evolución de la madurez del código fuente con explicaciones puntuales pero precisas, agregando diagramas generados a partir del código fuente y refleja un acabado impecable. (El profesor puede preguntar para refrendar calificación). | 4 | | | |
| Total | | 16 | | | |